

A380X MFD FMS and SURV pages to the FCOM

Every MFD FMS page laid out on its FCOM figure, the missing pages and functions

Branch	feature/a380/mcdu-improvements
Aircraft	A380X
Commits	8 own commits (on PR 6): 3800ff39d, e9d159981, ed33c1686, 752007449, fbef33990, edb07a695, 197a5f6aa, a09803fa8
Base	feature/a380x-checklists-surv
Status	In develop. Deployed and used by the user. PR draft 10.

Specification

- All 27 existing FMS pages audited against the A380 FCOM (DSC-22-FMS-20-30, PDF p.1009-1409): layouts rebuilt on the FCOM figures (absolute FCOM positions, border-box sizes as in Coherent GT), FCOM data entry formats (DSC-22-FMS-20-100).
- New pages: ALTERNATE, CLOSEST AIRPORTS, OFFSET, EQUI-TIME POINT, LL XING / TIME MKR, DATA / PRINTER, COMPANY WIND DATA REQUEST, COMPANY T.O DATA REQUEST, RECEIVED COMPANY T.O DATA, COMPANY F-PLN REPORT / REQUEST, REJECTED ATC INFO, TRANSFER TO MAILBOX.
- Functions: FMS print functions with printouts in the flypad; FUEL PLANNING computes and confirms the minimum BLOCK (FCOM P 179); PERF CRZ LRC; VERT REV RTA; FMS P/N panel; pilot-stored runways.
- SURV pages (CONTROLS, STATUS & SWITCHING) on DSC-34-20-60-50; SURV memos of the WXR and TAWS, TCAS STBY with ALT RPTG OFF.

Kept where FBW had a newer standard than the 2011 FCOM only when the user asked; the FCOM PDF is the reference for layout.

PR description (from the PR draft)

From `pr-drafts/10-a380x-mfd-fcom-pages.md`, section "Summary of Changes" onwards.

Summary of Changes

Every A380X MFD FMS page was compared with its figure in the A380 FCOM (DSC-22-FMS-20-30, PDF pages 1009-1409) and re-laid out on it, and the FCOM pages and functions FBW did not have were added. Each layout value and rule names its FCOM page in a code comment.

- **Layout:** the pages use the FCOM figure coordinates (`common/FcomLayout.tsx: fcomAt, fcomRight, fcomCentre, fcomLine`), the global FMS frame of the figures (title bar, 40 px entry fields, 22.5 px labels, 38 px tab bar), and box sizes counted border-box as Coherent GT renders them. Field contents are centred, dropdown arrows sit inside the frame, open menus are raised above the next items.
- **Entry formats:** the FCOM data entry formats (DSC-22-FMS-20-100) for every field, with the FCOM error messages.
- **New pages:** ALTERNATE, CLOSEST AIRPORTS, OFFSET, EQUI-TIME POINT, LL XING / TIME MKR, DATA / PRINTER, COMPANY WIND DATA REQUEST, COMPANY T.O DATA REQUEST, RECEIVED COMPANY T.O DATA (laid out; wired by PR 11), COMPANY F-PLN REQUEST / REPORT, REJECTED ATC INFO, TRANSFER TO MAILBOX.
- **Functions:** FMS print functions with the printouts shown in the flypad; FUEL PLANNING computes the minimum BLOCK from the predictions and CONFIRM BLOCK inserts it (FCOM P 179); PERF CRZ LRC (99 % of the maximum specific range, FCOM PER-IFT) and MAX TURB speed; VERT REV RTA with the time error thresholds; FMS P/N panel; pilot-stored runways; AIRWAYS with 31 segments and automatic airway connection; DATA / STATUS and AIRPORT behaviour per the FCOM.
- **SURV pages:** CONTROLS and STATUS & SWITCHING on DSC-34-20-60-50 P 4 / P 6 (XPDR AUTO / ON / STBY, TCAS and WXR option rules, ELEVN / TILT and GAIN fields); SURV memos of the WXR and TAWS; TCAS STBY with ALT RPTG OFF.

Where FBW had a newer FMS standard than this 2011 FCOM and the FCOM figure shows something else, the FCOM figure was followed.

Cockpit API Changes

New LVars documented in `fbw-a380x/docs/a380-simvars.md` for the SURV pages (WXR ELEVN / TILT / GAIN settings). No LVar removed.

Screenshots (if necessary)

TO ADD: each FMS page next to its FCOM figure (before / after).

References

- A380 FCOM DSC-22-FMS-20-30 (MFD FMS pages, PDF p.1009-1409), DSC-22-FMS-20-100 (data entry formats), DSC-34-20-60-50 (SURV pages).

Additional context

The pages were compared with the FCOM figures in a local browser harness that renders the MFD bundle with the sim's box model.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Open every FMS page (ACTIVE, POSITION, SEC INDEX, DATA menus) and compare it with its FCOM figure.
2. Enter values in each field, including out-of-range and badly formatted entries: FCOM format and error messages.

3. PRINT functions: the printout appears in the flypad.
4. FUEL PLANNING with a flight plan, cruise FL, ZFW and ZFWCG and no BLOCK: the computed BLOCK appears, CONFIRM BLOCK inserts it.
5. SURV CONTROLS / STATUS & SWITCHING: every button and field.